Breast Cancer Classification

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***Abstract*—In this project, we aim to classify histopathological images as benign or malignant using a fine-tuned version of the ResNet50 model or a trained from scratch model. For a better understanding of the results, we visualize the regions of the image that influence the classification process using Grad-CAM. On the BreaKHis 400X dataset with the fine-tuned model, we obtained a 98% accuracy and with the trained from scratch one an accuracy of 84%.**

I. Introduction

Breast cancer is one of the most common cancers worldwide, and early detection is crucial for improving patient outcomes.

A. Project Overview

This project was made with the purpose of classifying images of breast abnormal tissue. We compare transfer learning with training from scratch to understand the differences between the two approaches. The results are presented through a web application made with Streamlit that allows image upload, real-time classification and history of the result obtained in the last 8 images uploaded.

II. Dataset

In our work, we used the BreaKHis 400X dataset [1], consisting of histopathological images split into train and test sets. The train set contains 1148 images (371 benign, 777 malignant), further split 80/20 into train and validation. The test set contains 545 images (176 benign, 369 malignant). The dataset has, across both splits, approximately twice as many malignant images as benign.

III. Architecture

A. RestNet50

ResNet50 [3] is a deep convolutional neural network architecture pretrained on the ImageNet dataset, made of millions of images from different categories. Unlike other CNN architectures, ResNet50 introduces skip connections that allow gradients to flow more easily during training, solving the vanishing gradient problem. It is a good balance between performance and training speed compared to smaller (ResNet18) or larger (ResNet101) variants. On the last fully connected layer, it has 1000 neurons for classifying the output into one of the 1000 classes. We replaced the final fully connected layer with a new layer with 2 output neurons corresponding to the benign and malignant classes.

B. Transfer Learning

Transfer learning represents taking a pretrained model and fine-tuning it on a specific task. Instead of training from scratch with random weights, our fine-tuned model can already identify low-level features such as edges, corners and textures and build on them, learning high-level features.

IV. Training

For training, we used CrossEntropyLoss, a standard loss function for classification tasks, and the Adam optimizer [7] with a learning rate of 1e-4. To improve convergence, we added a ReduceLROnPlateau scheduler that reduces the learning rate by a factor of 10 when validation accuracy does not improve for 3 consecutive epochs, and also added early stopping with a patience of 7. Training was set to a maximum of 100 epochs, but the pretrained model stopped after 22 and the from scratch model after 15. Data augmentation was applied only on the train set, including horizontal and vertical flips, rotation up to 15 degrees, and brightness and contrast variations. The dataset was split 80/20 into train and validation, with the test set kept separate for final evaluation.

V. Results

A. Loss curves

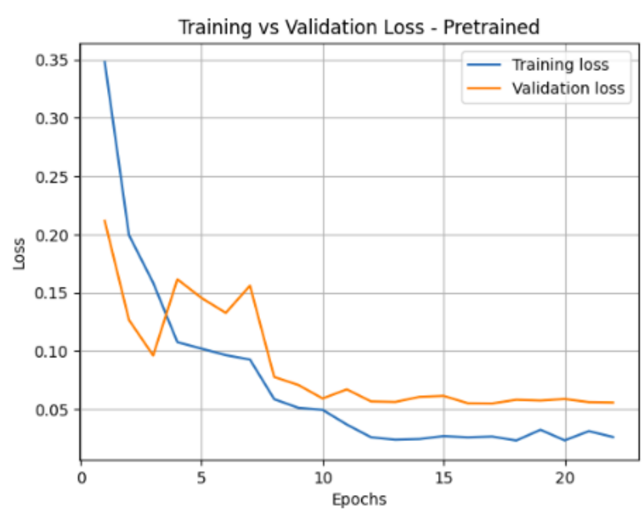

Fig. 1. Loss Curve - Pretrained Model

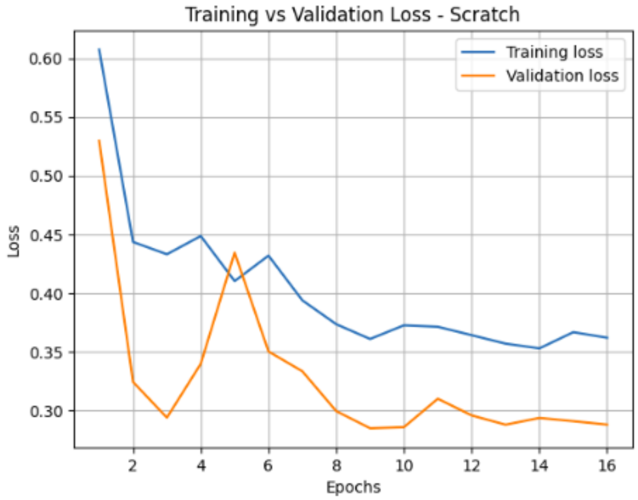

Fig. 2. Loss Curve - From Scratch Model

As seen in Figures 1 and 2, our pretrained model converges rapidly and remains stable throughout training, while the second model shows higher loss values and more oscillations, suggesting it struggles to generalize on this dataset.

B. Performance Metrics

TABLE I
MODEL PERFORMANCE ON TEST SET

Model	Class	Precision	Recall	F1
Pretrained	Benign	0.95	0.97	0.96
	Malignant	0.98	0.98	0.98
From Scratch	Benign	0.81	0.68	0.74
	Malignant	0.86	0.92	0.89

The pretrained model outperforms the from scratch model on both classes. This is most probably influenced by the class imbalance in the dataset, where malignant images are twice as frequent as benign ones.

C. Confusion Matrix

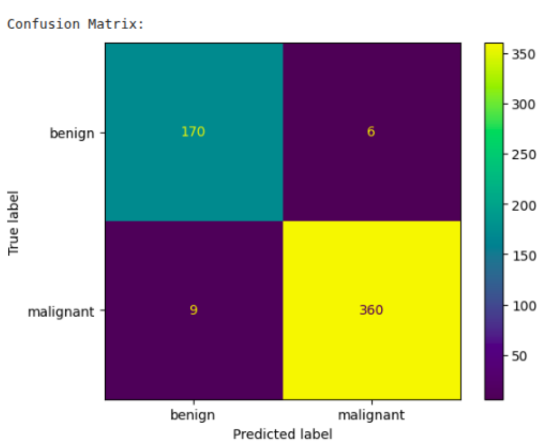

Fig. 3. Confusion Matrix - Pretrained Model

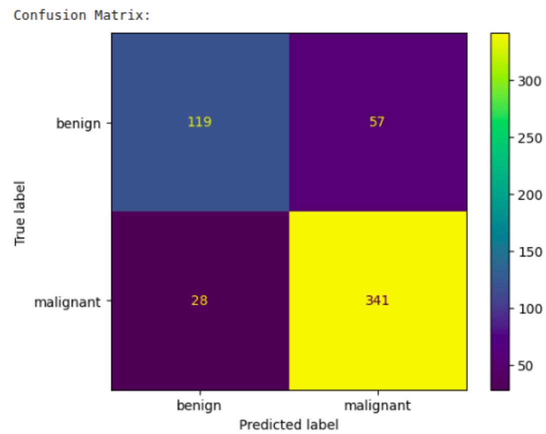

Fig. 4. Confusion Matrix - From Scratch Model

The pretrained model misclassified 15 images out of 545, while the from scratch model misclassified 85. The most significant difference is on the benign class, where the from scratch model misclassified 57 images as malignant compared to only 6 for the pretrained model.

VI. GRAD-CAM

Grad-CAM [2] (Gradient-weighted Class Activation Mapping) is a visualization technique that highlights the regions of the image that influenced the model's decision. We used it to better understand what the model focuses on when classifying an image as benign or malignant.

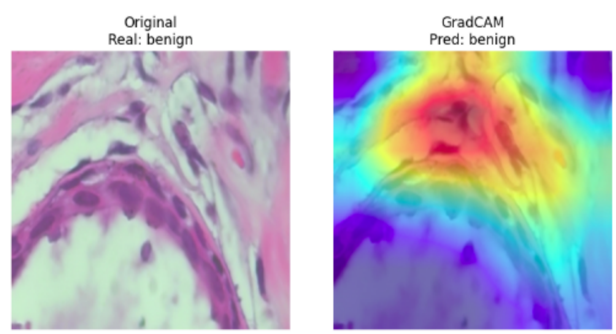

Fig. 5. Grad-CAM Visualization - Benign sample correctly classified

VII. APPLICATION

We built a web application using Streamlit [6] that allows users to upload a histopathological image and get a real-time classification from both models. The application displays the original image, the predicted class, the confidence score, and the Grad-CAM visualization for each model side by side. It also keeps a history of the last 8 uploaded images.

VIII. CONCLUSIONS

We compared a fine-tuned ResNet50 model with a model trained from scratch on the BreaKHis 400X dataset. The results confirm that transfer learning is more effective for this task, especially on the benign class. As future work, the model could be extended to support segmentation or detection tasks.

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